

Mechanistic parameters

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1 Introduction

The BLRPR_x model is defined by the following parameters: $\{\lambda, \iota, \theta, \alpha, \nu, \kappa, \phi\}$ (Kaczmarska et al. (2014); Onof & Wang (2020); Kim & Onof (2020)). These are, respectively:

- the storm arrival rate (hrs^{-1});
- the mean cell intensity scaling factor (dimensionless): $E(X|\eta) = \iota\eta$;
- the mean cell squared intensity scaling factor (dimensionless): $E(X^2|\eta) = \theta\eta^2$;
- the shape parameter of the gamma distribution of η (dimensionless);
- the rate parameter (inverse of the shape parameter) of the gamma distribution of η (hrs);
- the mean rate of cell arrival scaling factor: $\beta = \kappa\eta$ (dimensionless);
- the duration of storm activity parameter scaling factor: $\gamma = \phi\eta$ (dimensionless).

where η is the storm relative parameter of the exponential distribution of cell durations, β is the storm relative rate of arrival of cells and γ is the storm relative parameter of the exponential distribution of durations of storm activity. Each storm thus has a different value of these 3 storm relative parameters while the scaling factors defined above are the constant model parameters.

The parameterisation $\{\lambda, \iota, \theta, \alpha, \nu, \kappa, \phi\}$ can also usefully be replaced by $\{\lambda, \iota, f_1, \alpha, \nu, \kappa, \phi\}$ where $\theta = f_1\iota^2$. In other words, f_1 is defined by:

$$E(X^2|\eta) = f_1[E(X|\eta)]^2$$

Most of these parameters do not have straightforward physical/phenomenological interpretation. It is therefore useful to define a set of functions of these parameters which do possess such interpretability. These are called *mechanistic parameters*.

2 Mechanistic parameters

2.1 Rate of storm arrival

The only original parameter that is easily interpretable is the rate λ of arrival of storms. Note that model storms are not to be identified with physical storms. Although the model obviously aims at reproducing the clustering of cells in storms, it uses no definition of what might count as a storm in time-series of rainfall at a given time-step. This is a topic worthy of further investigation.

2.2 Mean cell intensity

This is obtained by taking the mean over the distribution of η of the storm relative mean:

$$\begin{aligned} E(X) &= E_\eta[E(X|\eta)] \\ &= E_\eta[\iota\eta] \\ &= \iota \frac{\alpha}{\nu} \end{aligned}$$

Although cells overlap one another in the model, the mean cell intensity is an important indicator of the overall intensity of the precipitation.

2.3 Mean cell duration

This is obtained by taking the mean over the distribution of η of the storm relative mean cell duration. Since this storm relative cell duration is exponentially distributed with parameter η , its mean is η so:

$$\begin{aligned} E(D_c) &= E_\eta[E(D_c|\eta)] \\ E(D_c) &= E_\eta[1/\eta] \\ &= \int_0^\infty \frac{\nu^\alpha t^{\alpha-2} e^{-\nu t}}{\Gamma(\alpha)} dt \\ &= \frac{\nu}{\alpha-1} \int_0^\infty \frac{\nu^{\alpha-1} t^{\alpha-2} e^{-\nu t}}{\Gamma(\alpha-1)} dt \\ &= \frac{\nu}{\alpha-1} \end{aligned}$$

2.4 Mean number of cells per storm

The way the current model has been specified, each storm starts with a cell (there are therefore no empty storms). We can show that the distribution of the number of cells after that first cell, is a gamma distribution with mean κ/ϕ .

To do that, consider that the probability of k cells in a storm (after the first fixed cell) can be obtained by taking the mean value over the distribution of η of the probability of k cells relative to a given storm (defined by η):

$$P(k \text{ cells}) = E_\eta[P(k \text{ cells}|\eta)]$$

To obtain $P(k \text{ cells}|\eta)$, we use the fact that, if the active duration of the storm (time during which cells can arrive) is t , then the distribution of the number of cells during that interval of duration t is Poisson distributed with parameter $\kappa\eta$. We then just have to integrate over the distribution of the possible values of t , i.e. the exponential distribution of parameter $\phi\eta$. We therefore obtain, for the overall probability of k cells (after the first fixed cell):

$$\begin{aligned}
P(k \text{ cells}) &= E_\eta \left[\int_0^\infty \phi \eta e^{-\phi \eta t} \frac{(\kappa \eta t)^k e^{-\kappa \eta t}}{k!} dt \right] \\
&= E_\eta \left[\frac{\phi \eta (\kappa \eta)^k}{k!} \int_0^\infty t^k e^{-(\kappa + \phi) \eta t} dt \right]
\end{aligned}$$

By doing k integration by parts, we find:

$$\begin{aligned}
P(k \text{ cells}) &= E_\eta \left[\frac{\phi \eta (\kappa \eta)^k}{[(\kappa + \phi) \eta]^k} \int_0^\infty t^k e^{-(\kappa + \phi) \eta t} dt \right] \\
&= E_\eta \left[\frac{\phi \eta (\kappa \eta)^k}{[(\kappa + \phi) \eta]^{k+1}} \right] \\
&= \frac{\phi \kappa^k}{(\kappa + \phi)^{k+1}} \\
&= \frac{\phi}{\kappa + \phi} \left[\frac{\kappa}{\kappa + \phi} \right]^k \\
&= \frac{\phi}{\kappa + \phi} \left[1 - \frac{\phi}{\kappa + \phi} \right]^k
\end{aligned}$$

This is the probability mass function of a geometric distribution with parameter $\frac{\phi}{\kappa + \phi}$.

Note that there are two ways of defining the geometric distribution; the one used here counts the number of failures before a success.

The mean μ_c number of cells can therefore easily be found as 1 plus the mean of this distribution:

$$\begin{aligned}
E(N_c) &= 1 + \frac{1 - \frac{\phi}{\kappa + \phi}}{\frac{\phi}{\kappa + \phi}}, \text{ i.e.} \\
E(N_c) &= 1 + \frac{\kappa}{\phi}
\end{aligned}$$

Since this mean is expressed in many papers as μ_c (e.g. Onof & Wheater (1993)), this can be rewritten:

$$\mu_c = 1 + \frac{\kappa}{\phi}$$

2.5 Mean duration of storm activity

The storm duration (i.e. from the beginning of the first cell, to the end of the last active cell) is only obtained as an approximation (see ???). What is easy to obtain, and therefore defines a plausible mechanistic parameter, is the mean time during which cells can arrive. This is the mean duration of storm activity. This is obtained by taking the mean over the distribution of η of the storm relative mean duration of storm activity. Since this storm relative duration of storm activity is exponentially distributed with parameter $\phi \eta$, its mean is $(\phi \eta)$ so:

$$\begin{aligned}
E(D_s) &= E_\eta [E(D_s | \eta)] \\
E(D_s) &= E_\eta [1/(\phi \eta)] \\
&= \frac{\nu}{\phi (\alpha - 1)}
\end{aligned}$$

using the calculations done for the mean cell duration above.

2.6 Mean total storm depth

The total depth of the storm is obtained by summing the cell depths over all the cells in the storm. The cell depth, in turn, is the product of the cell intensity and the cell duration.

Since, relatively to a storm, the cell intensity and duration are independent random variables, the storm relative mean contribution of a cell to the rainfall depth is:

$$\begin{aligned} E(R_c|\eta) &= E(X|\eta)E(D_c|\eta) \\ &= \iota\eta\frac{1}{\eta} \\ &= \iota \end{aligned}$$

Note that, since this is independent of η , ι is also the mean contribution to the total depth of a single rainfall cell. This provides us with an interpretation of parameter ι :

$$E(R_c) = \iota$$

For our purposes here, i.e. estimating the mean total storm depth, given that the cell depth and intensity are independent of the process of arrival of cells (which depends only upon the cell arrival rate and the storm duration parameter), the mean total storm depth is:

$$\begin{aligned} E(R_s) &= E_\eta(E(N_c|\eta)E(R_c|\eta)) \\ E(R_s) &= E(N_c)E(R_c), \text{ i.e.} \\ E(R_s) &= \iota(1 + \kappa/\phi) \end{aligned}$$

2.7 Other mechanistic parameters

If we wanted to completely reparameterise the model with mechanistic parameters, we would need one more parameter. If we choose an exponential distribution for the cell intensity, then

$$f_1 = 2$$

which means that $\theta = 2\iota^2$, we have enough mechanistic parameters.

3 Mechanistic model parameterisation

3.1 Exponential conditional cell intensity distribution

Assuming first that we use an exponential distribution to represent the cell intensity conditional upon η , we therefore have, as alternative parameterisation, the following set:

$$\{\lambda, E(X), E(D_c), \mu_c, E(R_s), E(D_s)\}$$

For convenience, these can be given new names, so that the parameter set (in the same order) is now:

$$\{\lambda, C_R, C_D, \mu_c, S_R, S_D\}$$

To reparameterise the model, all that is needed are the relations defining the original parameters in terms of the mechanistic ones. These relations, together with the original model equations, define the new model equations in terms of mechanistic parameters.

These relations are:

$$\begin{aligned}\iota &= S_R/\mu_c \\ \alpha &= \frac{\mu_c C_R C_D}{\mu_c C_R C_D - S_R} \\ \nu &= \frac{C_D S_R}{\mu_c C_R C_D - S_R} \\ \kappa &= (\mu_c - 1) \frac{C_D}{S_D} \\ \phi &= \frac{C_D}{S_D}\end{aligned}$$

3.2 General two parameter conditional cell intensity distribution

For the time being, we can just add parameter f_1 to the mechanistic parameter set, This parameter is the coefficient of variation of the conditional cell intensity distribution, i.e. the distribution of $X|\eta$. It is therefore a *coefficient of storm internal cell intensity variability*. The set of mechanistic parameters is therefore now:

$$\{\lambda, C_R, f_1, C_D, \mu_c, S_R, S_D\}$$

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